



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL
SENIOR CERTIFICATE

GRADE 12

ENGINEERING GRAPHICS AND DESIGN P1

NOVEMBER 2022

MARKS: 100

TIME: 3 hours

This question paper consists of 6 pages.

Barcode label

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.
2. Answer ALL the questions.
3. ALL drawings are in first-angle orthographic projection, unless otherwise stated.
4. ALL drawings must be prepared using pencil and instruments, unless otherwise stated.
5. ALL answers must be drawn accurately and neatly.
6. ALL the questions must be answered on the QUESTION PAPER, as instructed.
7. ALL the pages, irrespective of whether the question was attempted or not, must be re-stapled in numerical sequence in the TOP LEFT-HAND CORNER ONLY.
8. Time management is essential in order to complete all the questions.
9. Print your examination number in the block provided on every page.
10. Any details or dimensions not given must be assumed in good proportion.

FOR OFFICIAL USE ONLY															
QUESTION	MARKS OBTAINED			$\frac{1}{2}$	SIGN	MODERATED			$\frac{1}{2}$	SIGN	RE-MARKING			$\frac{1}{2}$	SIGN
1															
2															
3															
4															
TOTAL															
	2	0	0			2	0	0			2	0	0		

FINAL CONVERTED MARK	CHECKED BY
100	

COMPLETE THE FOLLOWING:
CENTRE NUMBER
CENTRE NUMBER
EXAMINATION NUMBER
EXAMINATION NUMBER

STAPLE

LAND SURVEYOR'S CERTIFICATE OF THE CORNER HEIGHTS AND BOUNDARY LENGTHS OF STAND 3414

CORNER HEIGHTS IN METRES		BOUNDARY LENGTHS IN MILLIMETRES	
A	1600	AB	24230
B	1600	BC	8370
C	1599	CD	18380
D	1599	DE	31200
E	1601	EA	32400

SYMBOL LEGEND

	EVERGREEN TREES
	BUSHES
	2000 HIGH SECURITY FENCE
	1800 HIGH PALLISADE FENCE

NOTE:
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ARCHITECT'S SIGNATURE

CLIENT'S SIGNATURE

ANSWER 18
In the space below, draw, in neat freehand, the SANS 10143 hatching (symbol) for a (18.1) FACE BRICK WALL, and graphical symbol for a (18.2) STAIRCASE.

18.1

18.2

2	07/01/22	ADD WATER METER	USA B
1	06/01/22	COLOUR NEW ADDITION RED	USA B
REVISION	DATE	DESCRIPTION	DRAWN
KOMBELLA, JOUBERT & MINNAAR ARCHITECTURAL DESIGN 365 BORCHARDS AVENUE ERASMIA 0183 012 685 1857 www.kjmdesign.co.za			
PRINTED BY: STEVENS PRINTERS		DATE OF PRINT: 08/01/2022	
DRAWING TITLE: SITE PLAN			
PROJECT: PROPOSED NEW ADDITION FOR MR N KEENAN ON STAND 3414, 21 DOMBEYA STREET, HARRISMITH			
PROJECT NUMBER: FTW2022-59		DRAWING NUMBER: 1 OF 5	
DATE: 05/01/2022	DRAWN: CASPER	CHECKED: ANNIE Q	SCALE: 1 : 350
REFERENCE CODE: K1N-2022			

QUESTION 1: ANALYTICAL (CIVIL)

Given:
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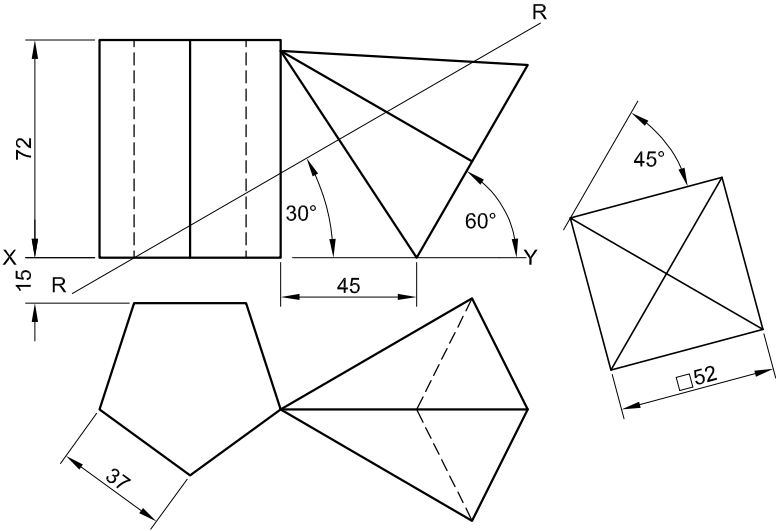
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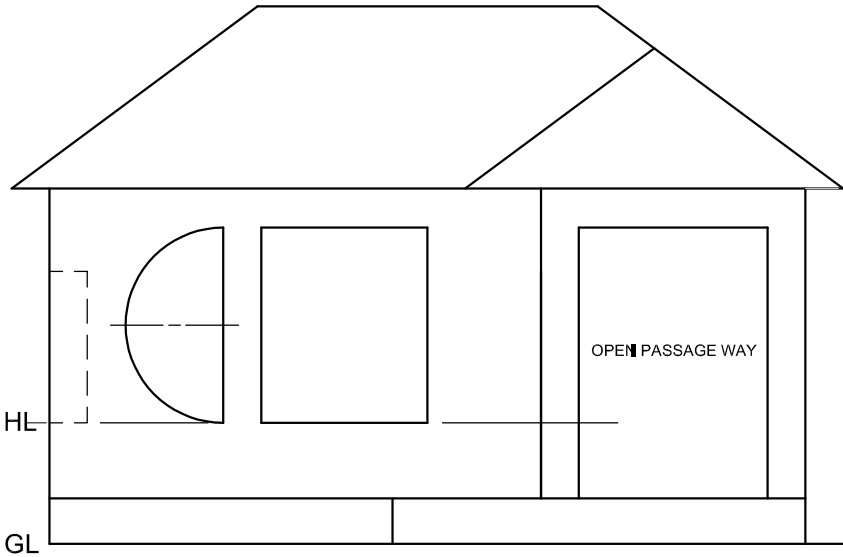
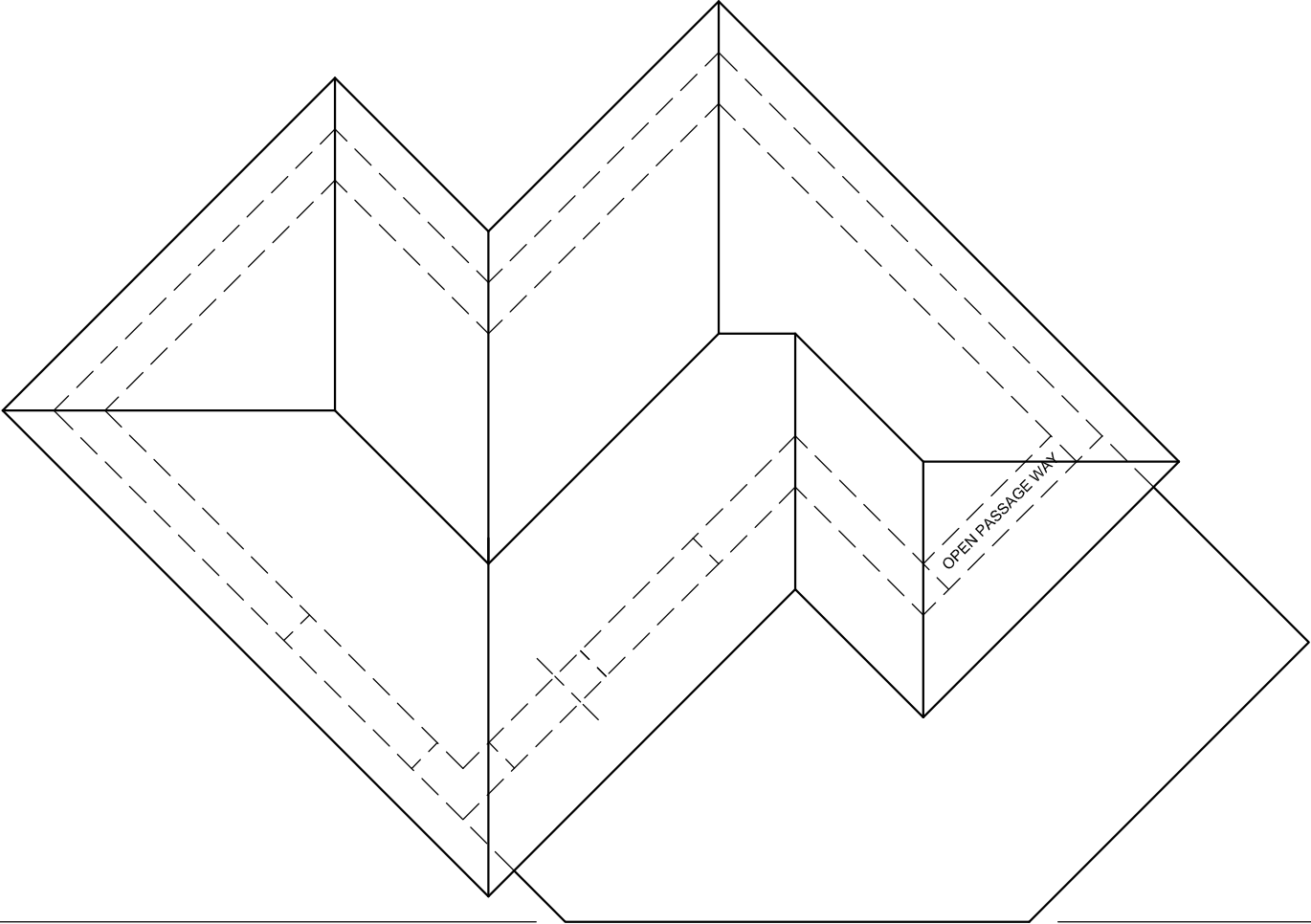
QUESTION 2: SOLID GEOMETRY

- Given:**
- The front view and top view of a right square pyramid that rests against a right regular pentagonal prism
 - An auxiliary view of the right square pyramid
 - Cutting plane R-R

- Instructions:**
- Draw, to scale 1 : 1, the following views of both solids:
- 2.1 The given front view
 - 2.2 A sectional top view with the parts above cutting plane R-R removed
 - 2.3 A sectional right view with the parts below cutting plane R-R removed
 - 2.4 The true shape of the cut surfaces of BOTH solids
- Planning is essential.
 - Show ALL hidden detail.
 - Show ALL construction.
- [37]



ASSESSMENT CRITERIA					
1	FRONT VIEW	7			
2	SECTIONAL TOP VIEW	12 ¹ / ₂			
3	SECTIONAL RIGHT VIEW	10			
4	TRUE SHAPE	6 ¹ / ₂			
5	CORRECT HATCHING	1			
PENALTIES (-)					
TOTAL		37			
EXAMINATION NUMBER					
EXAMINATION NUMBER					3



SP

QUESTION 3: PERSPECTIVE

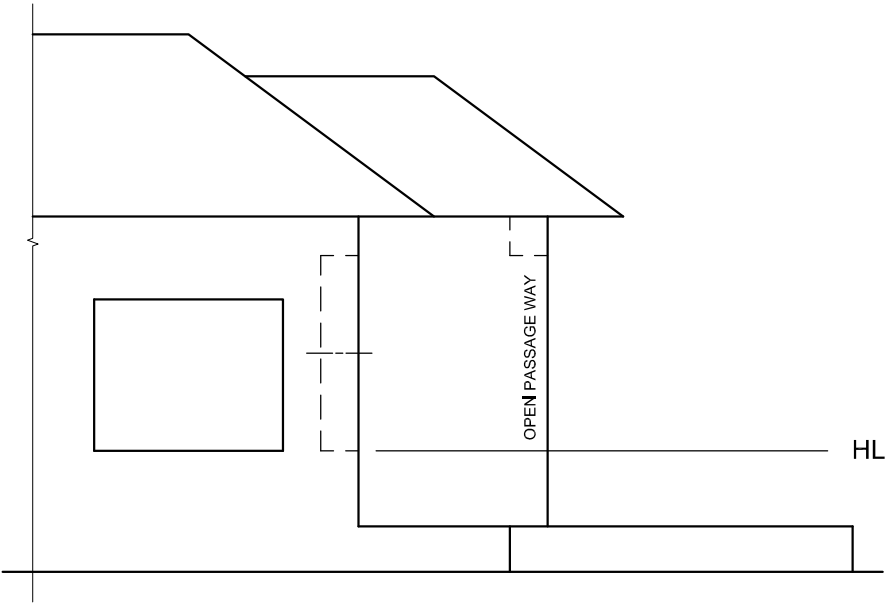
Given:
Three views of a club house and the information needed to draw a two-point perspective drawing
PP – Picture plane
HL – Horizon line
GL – Ground line
SP – Station point

Instructions:
Complete the perspective drawing.

- Align the drawing sheet with the ground line (GL).
- Determine and label the vanishing points.
- Show ALL construction.
- Show depth at the windows.
- NO hidden detail is required.

[41]

ASSESSMENT CRITERIA					
1	CONSTRUCTION	6			
2	WALL + BASE	10 ½			
3	WINDOWS + OPENING	10			
4	ROOF	7 ½			
5	ARC	7			
PENALTIES (→)					
TOTAL		41			



EXAMINATION NUMBER	
EXAMINATION NUMBER	4

WINDOW OPENING
ROOF CAP
RIDGE COVER
150 x 100 mm GUTTER ON
300 x 20 mm FASCIA BOARD
DOUBLE SWING DOOR OPENING
FINISHED FLOOR LEVEL
250 x 150 mm STEP
GROUND LEVEL
Ø100 mm RWDP THAT STOPS 100 mm ABOVE GROUND LEVEL

INCOMPLETE SOUTH ELEVATION

ROOM DESIGNATIONS

CHIMNEY DETAIL

700
45°
1200
1100
3700
GL

FLOOR FINISHES

LAUNDRY: TILES
BATHROOM: TILES
STUDY: CARPET
KITCHEN: TILES
BEDROOM: CARPET
LIVING ROOM: WOOD

INCOMPLETE FLOOR PLAN

FEATURES

D1 DOOR
D2 DOUBLE SWING DOOR
W1 WINDOW
W2 WINDOW
W3 WINDOW

FIXTURES

WC TOILET
WB WASH BASIN
WT WASH TUB
S SINK
B BATH

ELECTRICAL FITTINGS

1. ONE-WAY SWITCH - SINGLE-POLE
2. ONE-WAY SWITCH - DOUBLE-POLE
3. FLUORESCENT LIGHT 2 x 40 W
4. CEILING LIGHT
5. WALL-MOUNTED LIGHT
6. SWITCHED SOCKET OUTLET

NOTE:
THE ARROW SHOWS THE LIGHT CONNECTION TO THE SWITCH.

DOOR AND WINDOW SCHEDULE

TO FIT 50 TO FIT 50 2000
EQUAL EQUAL
2050 2050 1500
DOOR (D1) DOUBLE SWING DOOR (D2) WINDOW (W1)
600
1200 450 900
900 600
WINDOW (W2) WINDOW (W3)

WINDOW NOTES:

- A = OPENING SIDE
- B = FIXED PANEL
- ALL FRAMES = 50 mm
- 150 x 20 mm FIBRE CEMENT SILL UNDER ALL WINDOWS

ELECTRICAL SYMBOLS

ROOF COMPONENTS

	300 x 20 mm FASCIA BOARD
	150 x 100 mm GUTTER
	ROOF CAP AND RIDGE COVER

ROOF NOTES:
20° ROOF PITCH

114 x 40 mm ROOF TRUSSES ON
114 x 40 mm WALL PLATES

300 mm ROOF OVERHANG TO END OF ROOF TRUSS

20 mm CORRUGATED ROOF SHEETING ON
75 x 50 mm PURLINS @ 900 mm c/c

300 x 20 mm FASCIA BOARDS WITH
150 x 100 mm GUTTERS ON ALL SIDES

10 mm CEILING BOARD ON 40 x 40 mm
BRANDING STRIPS @ 540 mm c/c

INCOMPLETE FOUNDATION AND EXTERNAL WALL DETAIL

FIXTURES

500 150 800
FLOOR LEVEL
350
450 800 460
1200 400 200
850
1700 370 700
750 450 450 600 400
WASH BASIN (WB) TOILET (WC) SINK (S) BATH (B) WASH TUB (WT)

QUESTION 4: CIVIL DRAWING

- Given:**
- The incomplete south elevation of a new house, showing the walls, window and double swing door openings, step, position of the chimney, roof and labels
 - The incomplete floor plan showing the walls, step, positions of the doors, windows, chimney and fixtures, as well as electrical layout
 - An incomplete schematic diagram of a roof truss and roof notes
 - The incomplete foundation and external wall detail
 - Room designations and floor finishes
 - A table with chimney detail
 - A door and window schedule
 - A table of electrical symbols
 - A table of fixtures
 - The incomplete floor plan and position of the ground level of the new house, drawn to scale 1 : 50, and the incomplete foundation and a break line for the detailed section, drawn to scale 1 : 20, on page 6.

Instructions: Answer this question on page 6.

4.1 Using the given incomplete floor plan and ground level, draw, to scale 1 : 50, the following views of the new house:

4.1.1 THE COMPLETE FLOOR PLAN

Add the following features to the drawing:

- ALL doors and windows
- ALL fixtures as indicated by the abbreviations
- ALL electrical fittings as indicated by the numbers
- ALL the roof lines
- ALL hatching detail

4.1.2 THE COMPLETE SOUTH ELEVATION

Show the following features on the drawing:

- The outside walls, step, chimney, window and double swing door detail
- The roof detail, including the fascia boards, gutters and rainwater down-pipe
- The finished floor level

4.2 Using the given incomplete foundation and break line on page 6, draw, to scale 1 : 20, a DETAILED SECTION on cutting plane A-A of the area in the ellipse shown on the incomplete floor plan.

Show the following features on the drawing:

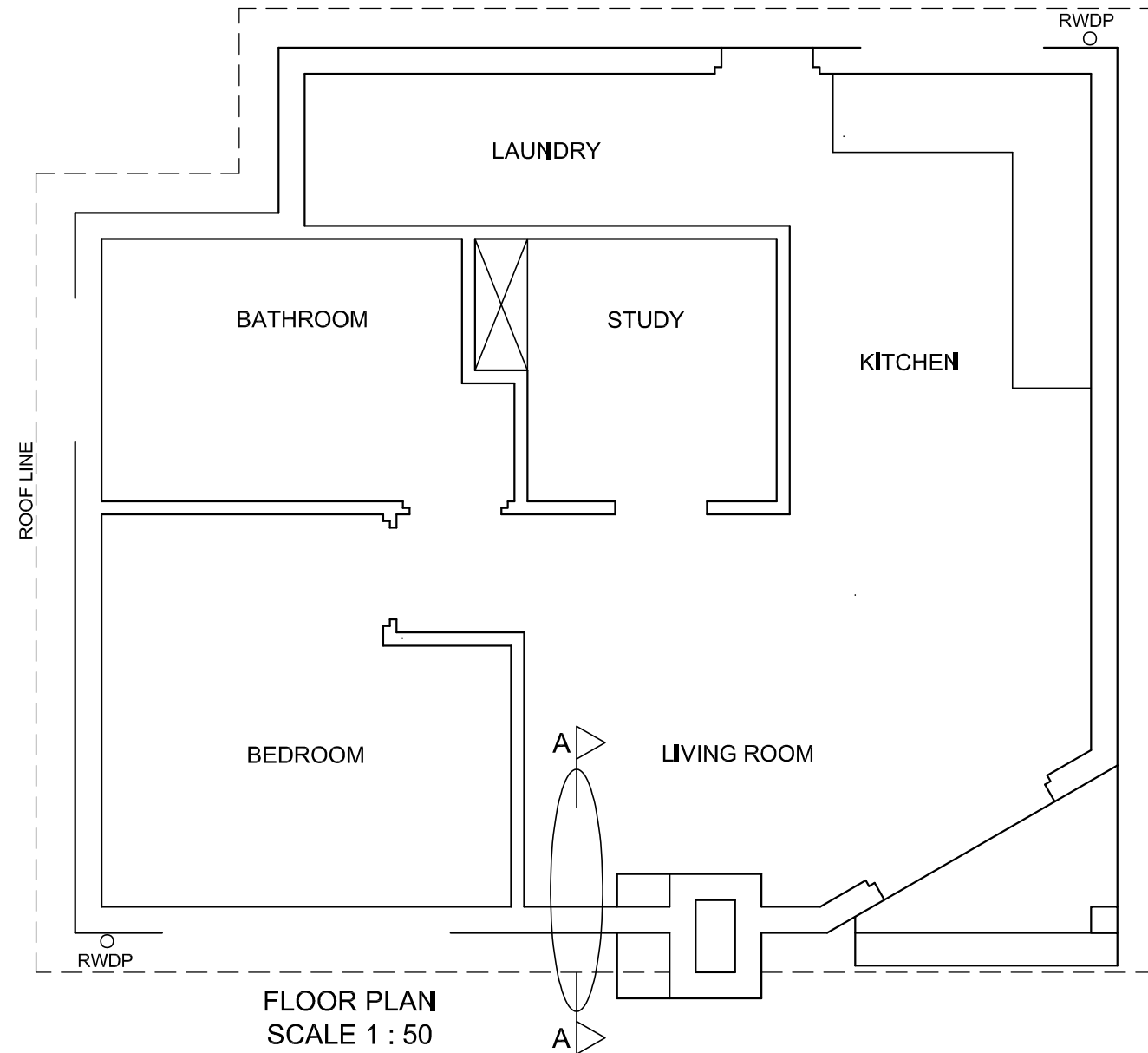
- The complete foundation and external wall detail
- The roof detail, including the fascia board and gutter
- The chimney and the double swing door and frame to the right (east) of cutting plane A-A
- ALL hatching detail. ONLY the substructure hatching may be drawn in neat freehand.

Label the following:

- The south elevation
- The floor finishes
- Ground level, finished floor level and damp-proof course (use the correct abbreviation and show it on ALL relevant views).

NOTE:

All drawings must comply with the guidelines and graphical symbols as contained in the SANS 10143.

GL

SECTION A-A
SCALE 1 : 20

MARK ALLOCATION FOR SECTION OF ROOF		FOR OFFICIAL USE ONLY	
A		INCORRECT SCALE(S) USED	
B		NON-ALIGNMENT OF VIEWS	
C		VIEW(S) ROTATED	
D		SECTION VIEWED INCORRECTLY	
E		INCORRECT LETTERING	
F			
G			
H			
TOTAL		TOTAL	

ASSESSMENT CRITERIA					
FLOOR PLAN					
		POSSIBLE	OBTAINED	SIGN	MODERATED
1	DOORS + WINDOWS	11			
2	FIXTURES + ROOF LINES	15			
3	ELECTRICAL	10			
4	HATCHING	3			
5	LABELS	3			
SUBTOTAL		42			
SOUTH ELEVATION					
1	ROOF + RWDP	8 $\frac{1}{2}$			
2	WALLS + STEP + CHIMNEY + FFL	7			
3	DOOR + WINDOW	7			
4	LABELS	1			
SUBTOTAL		23 $\frac{1}{2}$			
DETAILED SECTION					
1	ROOF DETAIL	11 $\frac{1}{2}$			
2	SLAB + WALL + CHIMNEY	8			
3	HATCHING	3 $\frac{1}{2}$			
4	DOOR	2 $\frac{1}{2}$			
5	LABELS	1			
SUBTOTAL		26 $\frac{1}{2}$			
TOTAL		92			
PENALTIES (-)					
GRAND TOTAL					
EXAMINATION NUMBER					
EXAMINATION NUMBER					6